

Guandi Wang

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EDUCATION

- **EURECOM** Sophia Antipolis, France
M.Sc. in Sensing Big Data for Intelligent Robots July 2026 – Present
- **KTH Royal Institute of Technology** Stockholm, Sweden
M.Sc. in ICT Innovation: Autonomous Systems and Intelligent Robots Aug. 2025 – Present
- Selected Coursework: Machine Learning, Distributed Artificial Intelligence, Introduction to Robotics, Hybrid and Embedded Control Systems.
- **Xi'an University of Architecture and Technology** Shaanxi, China
B.Sc. in Building Electrical and Intelligence Sep. 2021 – July 2025
- GPA: 83.5/100 (Top 10%) | **Technical Focus:** Control Theory, Power Engineering, Building Environment, Embedded Systems.

RESEARCH EXPERIENCE

Research Intern Sep. 2025 – Present
Tekniska Mekanik, KTH Royal Institute of Technology Supervisor: Junle Liu

ForeWind: Physics-Informed Spatiotemporal Wind Flow Forecasting

Constructed a 26-year hourly ERA5 benchmark (1999–2024) for wind forecasting over heterogeneous terrain. Built a physics-informed feature space from mean wind flow, wind variability, diurnal temperature range, elevation, and terrain roughness; used a Self-Organizing Map to identify nine wind regimes and select 12 representative coastal, inland, and plateau cities.

Developed a unified data and training pipeline for eight deep-learning architectures on city-centered 64×64 patches under a 24-hour input-to-24-hour forecast protocol, yielding 96 city-model configurations on Slurm-based HPC systems. Evaluated city RMSE, temporal correlation, patch-scale anomaly correlation, and persistence-normalized skill.

Showed that lower-boundary setting governs attainable error scale while architecture controls regime-specific failure modes. Two-way fixed-effect decomposition attributed most RMSE variability to city setting and persistence-normalized skill to city-model interaction; grid-level skill maps, lead-dependent errors, and power spectral density analysis exposed coherent variance attenuation across coastal and plateau regimes. Code and data.

Physics-Constrained Generative Reconstruction of Incompressible Flows

Investigating how generative flow-reconstruction models can strictly satisfy incompressibility through divergence-free constraints while preserving high reconstruction accuracy and calibrated uncertainty quantification.

Combining Gaussian Splatting with diffusion models to generate physically consistent flow fields on datasets including the Johns Hopkins Turbulence Database (JHTDB) and Taylor-Green vortex (TGV) benchmarks.

CFD-Informed Reinforcement Learning Environments for UAVs

Building two Isaac Lab UAV simulation environments from CFD data: a time-varying wind-field scenario based on JHTDB and a built-environment scenario driven by CFD flow around a Y-shaped building.

Designed the simulation lab for HPC execution and integrated ParaView with ROS2 to support real-time visualization and video streaming of large-scale aerodynamic environments.

GPU-Accelerated OpenFOAM Deployment on HPC

Deploying and validating GPU-enabled OpenFOAM workflows on supercomputing infrastructure for scalable CFD simulation and downstream learning tasks.

RESEARCH EXPERIENCE (CONTINUED)

Deep Multimodal Object Detection <i>Collaboration with Junle Liu</i>	
Developed spatial-mask interaction and channel-competition mechanisms for robust RGB–thermal object detection under sensor misalignment and adverse conditions, with an emphasis on deployable multimodal perception for real robotic systems. Accepted to BMVC 2026.	
Humanoid Sensing and Perception Guest Student <i>Istituto Italiano di Tecnologia (IIT)</i>	June 2026 – Sep. 2026 <i>Supervisor: Maria Lombardi; PI: Lorenzo Natale</i>
Unified Visual Segmentation and Annotation Tool	
Developed a visual segmentation and annotation tool using SAM3.1 as the foundation model and integrating RTMPose and RAFT for pose-aware annotation and optical-flow propagation.	
Supported precise frame-by-frame annotation, automatic object tracking across video sequences, and manual correction for both live-action movies and animated content.	
Force-Aware Physical Interaction with the R1 Humanoid Robot	
Developed a teleoperation-based physical interaction setup in which a human and the R1 humanoid robot jointly grasp an interaction-force sensor, enabling synchronized motion and force exchange for studying human–robot mechanical interaction.	

SELECTED PROJECTS

Autonomous Mobile Robot Navigation System (Coursework)	Sep. 2025 – Dec. 2025
Architected a navigation stack for TurtleBot3 using ROS2 (Foxy) and Python. Migrated control logic from Finite State Machines to Behavior Trees for enhanced reactivity.	
Integrated AMCL for robust localization and tuned PID controllers for smooth trajectory tracking.	
Photovoltaic Panel Defect Detection Algorithm <i>Bachelor Thesis</i>	Jan. 2025 – Jun. 2025
Grand Prize , Undergraduate Graduation Project Competition, Shaanxi Automation Association	2025.

PUBLICATIONS

Deep Multimodal Object Detection via Spatial Mask Interaction and Channel Competition. Accepted to BMVC 2026. arXiv:2608.02092.
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TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB, LaTeX
Frameworks & Libraries: PyTorch, ROS2 (Foxy/Humble), OpenCV, Scikit-learn, Isaac Lab, YARP, LeRobot
Tools & Platforms: Linux (Ubuntu), Git, Docker, Gazebo, Slurm (HPC), OpenFOAM, ParaView, VR systems

SELECTED AWARDS

Outstanding Graduate , Xi'an University of Architecture and Technology	2025
Academic Scholarship , Class II, Awarded for 3 consecutive years	2021 – 2023